RAUDEL ALEJANDRO GÓMEZ MOLINA

Computer Scientist | Researcher

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EDUCATION

Bachelor's Degree in Computer Science

University of Havana – Cuba

Nov 2021 – Mar 2025

- **Distinction:** Summa Cum Laude and Scientific Merit Award.
- GPA: 4.84/5.00
- Core areas: Algorithms, Data Structures, Algebra, Discrete Mathematics, Databases, Distributed Systems, Programming Languages Theory, Artificial Intelligence.
- Thesis: *An Approximation to the Language of All Satisfiable Boolean Formulas*.

TEACHING EXPERIENCE

Professor of Data Structures and Algorithms

University of Havana

Apr 2022 – Present | Havana, Cuba

- Teach the Data Structures and Algorithms course.
- Previously Student Assistant (Apr 2022 – Mar 2025) in Data Structures and Algorithms and Algebra.
- Mentored undergraduate students; designed problem sets and ICPC-style exercises for Competitive Programming training.
- Designed and helped implement lectures and practical classes for the course programs.

RESEARCH & ACADEMIC PROJECTS

An Approximation to the Language of All Satisfiable Boolean Formulas

Undergraduate Thesis – University of Havana

2024 – 2025

- Links SAT (NP-Complete) with formal language theory to construct the language of all satisfiable Boolean formulas.
- Two strategies: (1) transduction of a Copy language variant, proving NP-Hardness of the word problem for closed formalisms; (2) a Range Concatenation Grammar recognizing this language, showing RCGs capture all NP problems as formal languages.

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Mobile & Web

Bizne MX

Sep 2025 – Present | Mexico City, Mexico

Stack: Dart, Flutter, Getx, Next.js, BFF

- Promoted from Mobile Developer (Sep 2025 – May 2026) to Senior Software Engineer (Jun 2026).
- Built Next.js web apps with a Backend for Frontend (BFF) architecture and integrated them into the mobile apps through a native shell.

Full Stack Developer

jobs2live

Sep 2023 – Sep 2025 | Madrid, Spain

Stack: TypeScript, React, Next.js, C#, .NET, PostgreSQL

- Designed an N-layer architecture with the CQRS pattern and built LinkedIn scraping bots.

Mobile Developer

Business Seed

Feb 2024 – Dec 2024 | Mexico City, Mexico

Stack: Dart, Flutter, Getx

- Built hybrid Android and iOS apps from scratch following Clean Architecture, shipping MVPs in short timeframes.

RESEARCH INTERESTS

- Algorithms & Computational Complexity
- Competitive Programming
- Artificial Intelligence
- Software Engineering & Architecture
- Programming Language Theory

ACHIEVEMENTS

- **Summa Cum Laude** – Bachelor's Degree in Computer Science, University of Havana (2025).
- **Scientific Merit Award** – Recognized for outstanding undergraduate thesis.
- **Silver Medal** – ICPC Caribbean Finals 2023.
- 12th place – ICPC Caribbean Finals 2022.
- 10th place – ICPC Caribbean Finals 2021.
- National Pre-selection in Mathematics Olympiad (2 years).

LANGUAGES

- **Spanish** – Native
- **English** – Technical reading proficiency

SELECTED PROJECTS

Travel Agency

Distributed Systems Course Project

2023

- React, Antd, .NET, MySQL, OData.
- Multi-agent distributed platform for tourists and travel agencies.

Teslo Shop

E-commerce platform

- React, Next.js, Tailwind CSS, PostgreSQL.
- Full e-commerce with authentication, payments and product management.

TECHNICAL SKILLS

Theory: Algorithms, Data Structures, Competitive Programming, Algorithmic Thinking, Problem Solving, Research

Programming: C/C++, Python, Go, Rust, TypeScript, C#, Dart

Frameworks: React, Next.js, Flutter

Tools: Git, Docker, PostgreSQL, LaTeX